LIGHT, WAVES & RADIATION

The Quick Version

The essentials of the full book in a fraction of the pages. Read the Prologue for the whole story, then for each chapter: your questions with one-line answers, the key ideas, and the summary. Page numbers (“p. 57”) and figure numbers point to the full textbook, where each idea is explained properly.

PROLOGUE

The Whole Story in Ten Minutes

Radio, light, color, heat, sunburn, X-rays: one story told at different wavelengths. This is the map. When you feel lost in a detail, come back and ask: which step am I on?

1

Light is a ripple in electric and magnetic fields

Charges are surrounded by an **electric field**, and moving charges also make a **magnetic field**. Shake a charge and the change in its field spreads outward at $c = 3.00 \times 10^8$ m/s. A changing electric field makes a magnetic field and vice versa, so the ripple sustains itself and needs no medium. That ripple is an **electromagnetic (EM) wave**, with $c = \lambda f$.

→ CHAPTER 1

2

The wave’s energy arrives in packets called photons

Light delivers energy in lumps called **photons**, each with $E = hf = hc/\lambda$. Two separate dials: *frequency* (color) sets the energy per photon; *amplitude* (brightness) sets how many photons arrive. Proof: dim UV knocks electrons out of metal, but blinding red light can’t, because no single red photon has enough energy. A 500 nm photon carries $1240/500 \approx 2.5$ eV.

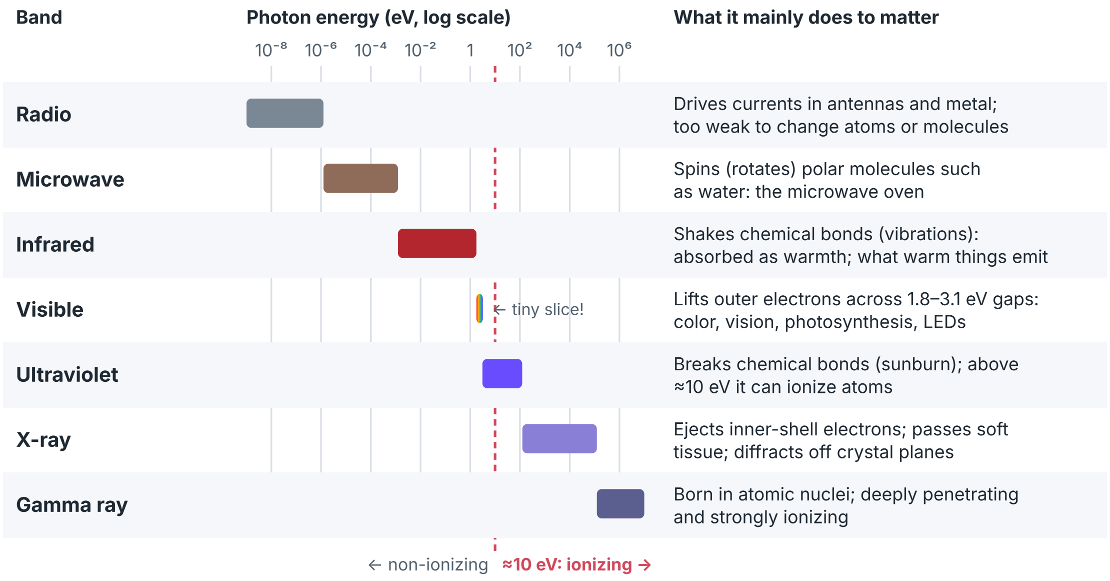
→ CHAPTER 2

3

Radio waves, light and X-rays are the same thing

From radio waves longer than buildings to gamma rays smaller than nuclei, it’s all the same thing traveling at c. The *only* difference is wavelength, which sets photon energy, which decides what the light does to matter (Figure P.1). Visible light (400–700 nm, 1.8–3.1 eV) is a tiny slice.

→ CHAPTER 3



Band	What it mainly does to matter
Radio	Drives currents in antennas and metal; too weak to change atoms or molecules
Microwave	Spins (rotates) polar molecules such as water: the microwave oven
Infrared	Shakes chemical bonds (vibrations): absorbed as warmth; what warm things emit
Visible	Lifts outer electrons across 1.8–3.1 eV gaps: color, vision, photosynthesis, LEDs
Ultraviolet	Breaks chemical bonds (sunburn); above ≈10 eV it can ionize atoms
X-ray	Ejects inner-shell electrons; passes soft tissue; diffracts off crystal planes
Gamma ray	Born in atomic nuclei; deeply penetrating and strongly ionizing

Wavelength runs the other way: about 1 km at the far left, about 10^{-13} m at the far right.

Figure P.1 The same phenomenon over 16 powers of ten. Each band's photon energy matches a different kind of "jump" in matter, which is why each band does something different. The red dashed line marks the rough ionizing threshold (about 10 eV).

4 Matter is electrons on an energy ladder

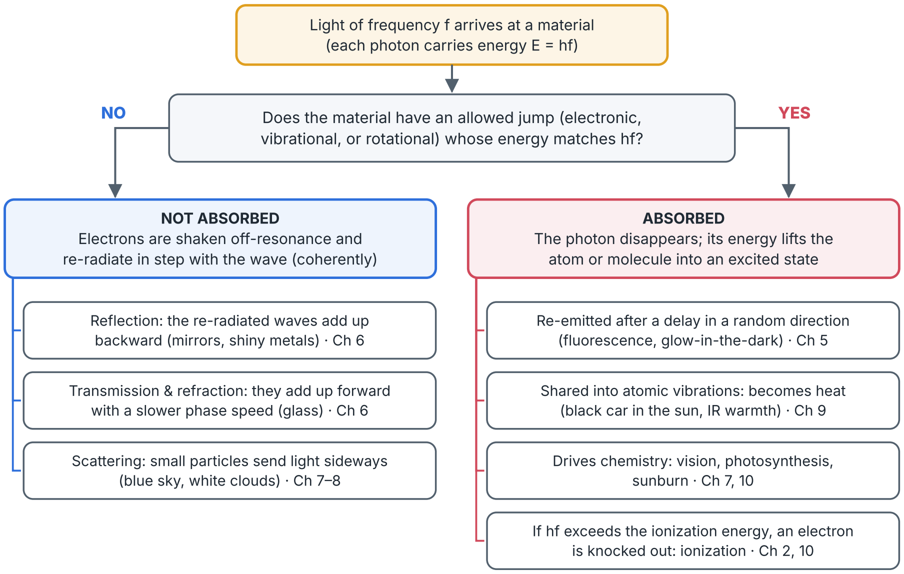
An atom is 0.1–0.3 nm across; its nucleus is tens of thousands of times smaller. Electrons live in probability clouds and can only have certain energies, like rungs on a ladder. Molecules add smaller rungs for *vibration* (infrared-sized) and *rotation* (microwave-sized). In solids the rungs merge into **energy bands** separated by **band gaps**.

→ CHAPTERS 4 AND 5

5 One question decides what light does to matter

Ask: *is there an allowed jump whose energy matches the photon's* hf*?* If yes, the photon is **absorbed**, and its energy is re-emitted, turned into heat, used for chemistry, or spent knocking out an electron. If no, the wave just shakes the electrons, and they re-radiate in step with it, which produces **reflection**, **transmission**, **refraction** and **scattering** (Figure P.2). Notice that the atom's *size* isn't in the question: an atom 5,000× smaller than a light wave absorbs it fine, because absorption is about matching *energy* (resonance).

→ CHAPTERS 5, 6 AND 8


Light of frequency f arrives at a material
(each photon carries energy E = hf)
Does the material have an allowed jump (electronic, vibrational, or rotational) whose energy matches hf?
NO
YES
NOT ABSORBED
Electrons are shaken off-resonance and re-radiate in step with the wave (coherently)
Reflection: the re-radiated waves add up backward (mirrors, shiny metals) · Ch 6
Transmission & refraction: they add up forward with a slower phase speed (glass) · Ch 6
Scattering: small particles send light sideways (blue sky, white clouds) · Ch 7–8
ABSORBED
The photon disappears; its energy lifts the atom or molecule into an excited state
Re-emitted after a delay in a random direction (fluorescence, glow-in-the-dark) · Ch 5
Shared into atomic vibrations: becomes heat (black car in the sun, IR warmth) · Ch 9
Drives chemistry: vision, photosynthesis, sunburn · Ch 7, 10
If hf exceeds the ionization energy, an electron is knocked out: ionization · Ch 2, 10

Most real materials do several of these at once; the mix depends on wavelength. Energy is conserved: $R + A + T = 1$.

Figure P.2 The master flowchart for light meeting matter. Almost every phenomenon in Part III is one of these boxes.

6 Color is what survives

A material absorbs the wavelengths that match its jumps; your eye gets what's left (a leaf absorbs red and blue, returns green). For solids: band gap above 3.1 eV, no visible absorption (pure diamond is clear); below 1.8 eV, all of it absorbed (silicon looks gray and opaque); in between, colored. The blue sky is different: air molecules scatter short wavelengths far more than long ones.

→ CHAPTER 7

7 Waves spread out and interfere

Light bends around obstacles and spreads through openings (**diffraction**), and overlapping waves reinforce or cancel (**interference**). So you can't image details smaller than about half a wavelength, which is why no light microscope shows atoms. Objects much smaller than the wavelength barely scatter it, which is why a microwave oven's door mesh blocks 12 cm microwaves but passes visible light.

→ CHAPTER 8

8 Everything glows, and warm things glow in infrared

Everything above absolute zero emits **thermal radiation** set by its temperature: you (310 K) peak near 9–10 μm in the infrared, a stove burner (~1000 K) glows dull red, the Sun (5800 K) peaks in the visible. IR is called "heat radiation" because warm everyday objects emit mostly IR and molecules absorb it straight into vibration. But IR is light, not heat, and visible light heats things too.

→ CHAPTER 9

9 Above about 10 eV, light becomes ionizing

A photon with more energy than an electron's binding energy (roughly 10 eV, far UV) can knock it off its atom, breaking bonds and damaging DNA. At everyday intensities, more brightness can't substitute for photon energy. "Radiation" also means fast *particles* (alpha, beta, neutrons, cosmic rays). Magnetic fields steer charged particles away, as Earth's magnetosphere does, but do nothing to photons; only matter stops those.

→ CHAPTER 10

10 X-rays fit between the atoms

X-ray wavelengths (~0.1 nm) match the spacing of atomic planes, so a crystal acts as a 3-D diffraction grating, reflecting only at angles that satisfy **Bragg's law**, $n\lambda = 2d\sin\theta$. That's how we know copper is FCC and iron is BCC. Your ME2340 Miller-index planes are literally the mirrors that reflect X-rays.

→ CHAPTER 11

KEY IDEA — **Three equations carry most of this book**

$c = \lambda f$ (wavelength and frequency are locked together) · $E = hf = \frac{hc}{\lambda}$ (photon energy grows with frequency) · $E_{\text{photon}} = \Delta E_{\text{jump}}$ (light is absorbed or emitted only when its photon energy matches an allowed jump). Nearly everything else is these three applied to a particular wavelength and material.

PART I · **WHAT LIGHT IS**

1 Electromagnetic Waves: Ripples in Fields

full chapter p. 12

"How does EM radiation actually work — what is physically waving?" p. 16

The electric and magnetic fields themselves are what wave: at each point the light passes through, the strength and direction of the electric field (and the magnetic field with it) swing back and forth, hundreds of trillions of times per second for visible light.

KEY IDEA

Accelerating charges make electromagnetic waves, and electromagnetic waves accelerate charges. Antennas, light bulbs, atoms and the Sun all emit by shaking charges; eyes, cameras and antennas all detect by being shaken. Light from atoms (Chapter 5) and from hot objects (Chapter 9) works the same way, with electrons as the wiggling charges.

SUMMARY

- An electric field is the force per unit charge at each point in space ($\vec{E} = \vec{F}/q$); moving charges also create a magnetic field $\vec{B}$.
- Accelerating charges launch kinks in their fields that travel at c. A steadily oscillating charge launches an EM wave; arriving waves shake charges in turn.
- Changing E makes B and changing B makes E, so the wave sustains itself and needs no medium. Its speed in vacuum is $c = 1/\sqrt{\mu_0\varepsilon_0} = 3.00 \times 10^8$ m/s.
- What waves is the field itself. E, B and the direction of travel are mutually perpendicular, and $B = E/c$.
- $c = \lambda f$: Wi-Fi at 2.4 GHz is 12.5 cm long; green light at 550 nm oscillates 5.45×10^{14} times per second.
- Intensity $I = \frac{1}{2}c\varepsilon_0 E_0^2$, so it grows as amplitude squared. Amplitude sets brightness; frequency sets color.
- Polarization is the direction of E. Polarizers pass one direction ($I = I_0\cos^2\theta$); polarized sunglasses block horizontally polarized glare.

2 Photons: Light as Packets of Energy

full chapter p. 22

"What is the photoelectric effect?" p. 23

The photoelectric effect is the ejection of electrons from a material, usually a metal, when light shines on it.

"Does a bigger amplitude of a wave mean more photons?" p. 28

Yes: for light of a given color, a bigger amplitude means more photons per second, while each photon keeps exactly the same energy, hf.

“What is a photon made up of?” p. 29

Nothing smaller: as far as every experiment can tell, a photon is an elementary particle with no internal parts.

KEY IDEA

Frequency sets the energy of each photon. Amplitude (brightness) sets how many photons arrive per second. Whether light can do something to a single atom or electron, such as eject it, break a bond or trigger your retina, depends on the energy per photon, not on the brightness.

KEY IDEA

Light travels like a wave and arrives like a particle. Use the wave picture for where light goes; use the photon picture for what light does when it is absorbed or emitted.

SUMMARY

- Light delivers energy in photons of $E = hf = hc/\lambda$. In practical units, $E\,(\mathrm{eV}) = 1240/\lambda\,(\mathrm{nm})$; visible photons carry 1.8–3.1 eV.
- Photoelectric effect: one photon frees one electron if $hf \geq \phi$, giving $KE_{\max} = hf - \phi$. Below the threshold $f_0 = \phi/h$, nothing happens at any brightness.
- The stopping voltage measures $KE_{\max} = eV_s$. Plots of $KE_{\max}$ vs. f are straight lines with slope h for every metal.
- Brightness is photon rate: $P = \dot{N}hf$. At fixed color, photon count $\propto$ amplitude²; a 5 mW green laser emits 1.3×10^{16} photons/s.
- A photon is an elementary quantum of the EM field: no mass, no charge, no parts; it has energy hf, momentum h/λ and spin.
- Light travels like a wave and is absorbed like a particle. Single photons build up interference patterns one dot at a time.

3 The Electromagnetic Spectrum

full chapter p. 32

“What is the EM spectrum — are radio waves, light and X-rays really the same thing?” p. 32

Yes. Radio waves, visible light and X-rays are exactly the same phenomenon: oscillating electric and magnetic fields traveling at *c*, delivered in photons of energy hf.

KEY IDEA

Photon energy, not brightness, decides what kind of interaction is possible. Below about 1 eV, light mostly makes things jiggle, which is heat. From about 1.8 to 10 eV it can rearrange outer electrons, which gives color and drives chemistry. Above about 10 eV it can rip electrons off atoms. Because absorption happens one photon at a time (Chapter 2), no number of low-energy photons adds up to one high-energy photon.

SUMMARY

- The EM spectrum is one phenomenon spread over more than 18 powers of ten in wavelength. Each band can be labeled by λ, $f = c/\lambda$ or $E = 1240\ \mathrm{eV} \cdot \mathrm{nm}/\lambda$.
- Order, long to short: radio (> 30 cm), microwave, infrared (700 nm–1 mm), visible (400–700 nm, 1.8–3.1 eV), ultraviolet (10–400 nm), X-ray (0.01–10 nm), gamma ray. Boundaries are conventions; X-ray and gamma overlap and are named by source.
- Each band is made by accelerating charge at a different scale: antennas, vibrating molecules, outer electrons, fast or inner-shell electrons, nuclei.
- Photon energy sets the interaction: heating below ~1 eV, electronic and chemical changes from ~1.8 to 10 eV, ionization above ~10 eV.
- The atmosphere passes visible/near-IR light and radio from ~1 cm to ~30 m; it blocks gamma, X-rays, most UV and most IR.

- Eyes use 400–700 nm because sunlight is plentiful there, air and water are clear, and those photon energies can flip retinal molecules without damaging them.

PART II · **ATOMS AND ENERGY LEVELS**

4 Inside the Atom: Scale, Orbitals, and Quantized Energy

full chapter p. 41

"How big is an atom, from one side to the other?" p. 42

About 0.1 to 0.5 nanometers, or one to five ten-billionths of a meter.

"If a marble is the nucleus and it's placed at the center of a football field, is the electron probability cloud the size of the football field?" p. 43

No: the cloud is bigger. With a 1.5 cm marble as the nucleus, a heavy atom such as gold would be about 300 m across, and a hydrogen atom nearly a kilometer.

KEY IDEA

An atom is nearly all empty of *mass*, but it is not empty of *electron*: the electron's cloud fills the whole volume. That cloud, not the nucleus, sets an atom's size, its chemistry, and how it interacts with light.

KEY IDEA

Confinement causes quantization. Any wave trapped in a limited space (a string, an organ pipe, an electron in an atom) can vibrate only in certain patterns, each with a definite energy. Free electrons, which are not trapped, can have any energy at all.

SUMMARY

- Atoms are about 0.1–0.5 nm across. Nuclei are femtometers across ($R \approx 1.2\ \text{fm}\ A^{1/3}$) and hold more than 99.9% of the mass.
- Scaled so the nucleus is a 1.5 cm marble, the electron cloud would be roughly 300 m (gold) to 940 m (hydrogen) across: bigger than a football field.
- An electron in an atom is a 3D standing wave (orbital). Its square is a probability cloud with no hard edge.
- Only whole-number standing-wave patterns fit ($n\lambda = 2\pi r_n$ in de Broglie's model), so bound electrons have only certain energies.
- Hydrogen: $E_n = -13.6\ \text{eV}/n^2$. The ground state is $n = 1$; 13.6 eV ionizes it.
- Ionization energies run from about 4 to 25 eV, all above visible photon energies.
- Matter is solid because overlapping electron clouds repel (electric repulsion plus Pauli exclusion).

5 Absorption and Emission: Electrons Jumping Up and Down

full chapter p. 51

"How can an atom that's thousands of times smaller than a light wave absorb it?" p. 52

Because absorption depends on matching the wave's *frequency*, not its size.

"When an electron jumps down an energy level, how does that make a photon, and what is 're-emission'?" p. 53

The photon is created during the jump itself; it is not stored inside the atom beforehand.

KEY IDEA

Light is absorbed by **resonance**: the photon's energy (its frequency) must match an energy gap. Brighter light delivers more photons, but never changes which gaps a photon can bridge.

KEY IDEA

Re-emission is not reflection. In reflection the wave shakes electrons that do *not* absorb it; they re-radiate in step with the wave and instantly, sending back one orderly beam at the mirror angle, same color. Re-emission follows a true absorption: delayed, random in direction and phase, sometimes a different color. Chapter 6 covers reflection.

SUMMARY

- One rule: $hf = E_{\text{upper}} - E_{\text{lower}}$. Absorption is resonance: the frequency must match, whatever the atom's size.
- A photon is created during a downward jump, as the mixed-state cloud sloshes at $f = \Delta E/h$. Re-emitted photons leave in random directions.
- Hydrogen: $\Delta E = 13.6\ \text{eV}(1/n_{\text{lower}}^2 - 1/n_{\text{upper}}^2)$; $3 \rightarrow 2$ gives the 656 nm red line.
- Hot dense objects: continuous spectra. Hot thin gases: emission lines. Cool gas before a hot source: dark lines at the same wavelengths.
- Absorbed energy is re-emitted, turned into heat, or used for chemistry. Reflection is coherent and involves no absorption.
- Fluorescence: absorb short, emit longer, within ns (Stokes shift). Phosphorescence: a stuck state releases light slowly.
- Vibrations absorb infrared (≈ 0.05–0.5 eV), rotations microwaves (≈ 0.001 eV), if they move charge.
- The band gap separates metals, semiconductors and insulators and sets an LED's photon energy. Lasers amplify by stimulated emission from a population inversion.

6 Reflection, Transmission, Absorption, and Refraction

full chapter p. 63

"How does light work? What actually reflects the wave when it hits something?" p. 65

The electrons in the material do: the incoming wave shakes them, and the waves they radiate add up to the reflected wave (and, going forward, to the transmitted wave).

"Is reflection the same thing as the material absorbing and re-emitting the EM wave?" p. 65

No. Reflection is coherent re-radiation by electrons that are *driven* by the wave; it is not a photon being absorbed into an excited state and re-emitted later.

KEY IDEA

R, A, and T depend on the material *and the wavelength*. Window glass transmits about 90% of visible light but almost none of the thermal infrared a warm object emits.

KEY IDEA

Crossing into a material changes light's speed and wavelength by the factor n, never its frequency. Bending (refraction), dispersion, lenses, and fibers all follow from that one fact.

SUMMARY

- Light meeting matter is reflected, absorbed, or transmitted (some scattered): $R + A + T = 1$, all wavelength-dependent.
- The wave drives electrons, which re-radiate at the same frequency and in fixed phase. Reflection and refraction are this coherent re-radiation, not absorb-and-re-emit (fluorescence), which is delayed, random in direction, and usually red-shifted.
- Roughness compared with the wavelength sets specular vs. diffuse reflection; $R = [(n_2 - n_1)/(n_2 + n_1)]^2$ gives about 4% per glass surface.
- Metals' free electrons respond below the plasma frequency: they reflect 90–95% and absorb the rest within about 0.1 µm. Glass passes visible light (gap far above 3.1 eV) but absorbs UVB and thermal IR.
- $n = c/v$; frequency is unchanged and wavelength becomes λ_0/n. Snell's law gives the bending; beyond $\sin\theta_c = n_2/n_1$, total internal reflection traps light in fibers.

- Dispersion separates colors in prisms and rainbows; lenses focus by refraction. Pores and birefringent grain boundaries scatter light, making ceramics translucent or opaque.

7 Color

full chapter p. 74

“How does color work in terms of EM radiation?” p. 75

Color is how your visual system labels the mix of visible wavelengths (photon energies of about 1.8 to 3.1 eV) entering your eye; an object's color is whatever part of the light falling on it the object sends toward you.

“What are the electron energy gaps that give things their color?” p. 77

They are the energy differences between states an electron can occupy, and a material is colored when one of its gaps falls between about 1.8 and 3.1 eV, the energies of visible photons.

KEY IDEA

Pigments don't add color; they take it away. Yellow plus cyan paint looks green because yellow removes blue and cyan removes red, so only green survives both.

KEY IDEA

An object doesn't "have" a color the way it has a mass. It has an absorption spectrum. The color you see is the illumination minus what the object absorbs, as read by your three cones.

SUMMARY

- Visible light is 400–700 nm, photon energies 3.1–1.8 eV. Color is the brain's reading of three cone signals (S, M, L peaks near 420, 534, 564 nm), so different spectra (metamers) can look identical.
- Light mixes additively (RGB screens); pigments mix subtractively (CMY inks), because each pigment removes part of the spectrum.
- Most objects are colored by what they absorb. The absorbed light becomes heat; the leftovers reach your eye.
- Band-gap rule: $E_g > 3.1$ eV is colorless, $E_g < 1.8$ eV is opaque and dark or metallic, and gaps in between are yellow to red. The edge is at $\lambda = 1240/E_g$ nm.
- Impurity ions (Cr^{3+} in ruby, Fe^{2+}–Ti^{4+} in sapphire) and conjugated molecules (dyes, carotene, chlorophyll) create visible-energy gaps. Gold and copper are colored by d-band absorption starting near 2.4 and 2 eV; silver's starts in the UV.
- Rayleigh scattering ($\propto 1/\lambda^4$) makes the sky blue and sunsets red; large cloud droplets scatter all colors and look white.
- Thin films and nanostructures make structural color by interference; emitters (neon, fireworks, LEDs, screens) show their emission spectra.

8 Interference and Diffraction (and Why Size vs. Wavelength Matters)

full chapter p. 84

“What is diffraction — and what does it mean in quantum physics?” p. 89

Diffraction is the spreading of a wave as it passes through an opening or around an edge, by an angle of roughly λ/a; it is strongest when the opening is about a wavelength wide (Section 8.2).

“What is the diffraction limit?” p. 91

The diffraction limit is the smallest detail any lens-based instrument can resolve, set by the wavelength: about $\lambda/(2\,\mathrm{NA})$, or roughly half a wavelength for the best microscopes.

“Why can big wavelengths interact with small objects?” p. 92

They can because a wave is an oscillating electric field, and anything containing charges feels it, whatever its size.

“Something so small (an atom) shouldn’t be able to interact with such a long wavelength — the wave should be unnoticeable to the tiny atom. What’s the idea here?” p. 93

Your intuition is right for *scattering and imaging*, but absorption and emission don’t depend on size at all; they depend on resonance.

KEY IDEA

Path difference decides everything. A whole number of wavelengths extra, $(0,\ \lambda,\ 2\lambda,\ \ldots)$, gives a bright spot; an extra half wavelength, $(\lambda/2,\ 3\lambda/2,\ \ldots)$, gives a dark one. The pattern only holds still if the waves keep a fixed phase relationship (they are **coherent**), which is why lasers make such clean patterns.

KEY IDEA

A wave squeezed through an opening of width a spreads by an angle of about λ/a. Diffraction is dramatic when the opening is about the size of the wavelength and negligible when the opening is thousands of wavelengths across.

KEY IDEA

You can’t resolve details much smaller than about half the wavelength you look with. To see smaller things, use a shorter wavelength or a bigger aperture.

SUMMARY

- Superposition: overlapping waves add. A path difference of $m\lambda$ gives a bright spot; $(m+\frac{1}{2})\lambda$ gives a dark one.
- Huygens’ principle explains diffraction. An opening of width a spreads a wave by about λ/a, which is dramatic when $a \sim \lambda$.
- Double slits and gratings: bright fringes where $d\sin\theta = m\lambda$. Gratings (and CDs) split white light into spectra; thin films color soap bubbles.
- Matter diffracts with $\lambda = h/p$. Particles sent one at a time still build an interference pattern, so they travel as waves and are detected as particles.
- The diffraction limit: angular resolution $1.22\lambda/D$; microscope resolution $\approx \lambda/(2\,\mathrm{NA}) \approx 200$ nm for visible light. Electrons and X-rays reach atomic scales.
- Size vs. wavelength: Rayleigh ($d \ll \lambda$, scattering $\propto d^6/\lambda^4$), Mie ($d \sim \lambda$), geometric ($d \gg \lambda$). This explains the microwave door and AM vs. FM.
- Atoms absorb and emit waves thousands of times larger than themselves because absorption is set by resonance, not size; small antennas are inefficient unless resonant.

9 Infrared, Heat, and Thermal Radiation

full chapter p. 96

“Is infrared the same thing as heat?” p. 100

No. Infrared is light, an electromagnetic wave just like visible light but with longer wavelengths (about 0.7 µm to 1 mm).

“Why are infrared waves often called ‘heat waves’?” p. 101

Because infrared is the radiation warm things give off and the radiation your skin feels as warmth.

“Why does infrared show heat?” p. 101

Because at everyday temperatures the light an object emits by itself is almost entirely infrared, and the amount rises steeply with temperature ($\propto T^4$).

KEY IDEA

Every object emits thermal radiation all the time. Double the absolute temperature and it emits 16 times the power, with its peak at half the wavelength. Whether something warms or cools depends on the *net* exchange with its surroundings.

KEY IDEA

Good absorbers are good emitters, and good reflectors are poor emitters (Kirchhoff's law). A thermal camera reads a surface's own glow only when its emissivity is high; on shiny metal it mostly sees reflections.

SUMMARY

- Thermal energy is random atomic motion, temperature measures its average, and heat is energy transferred by a temperature difference. Only radiation works through vacuum.
- Jiggling charges make every object emit a continuous thermal spectrum. Wien's law gives $\lambda_{\text{peak}} = 2898\ \mu\text{m} \cdot \text{K}/T$; Stefan–Boltzmann gives $P = \varepsilon\sigma A T^4$. You peak near 9–10 µm and radiate about 100 W net.
- Visible glow starts near 800 K (dull red), then goes orange, yellow, and white. Bulbs are labeled by color temperature, and "warm" light means a lower temperature.
- Infrared is light, not heat. It is called heat radiation because everyday objects emit it and molecular vibrations absorb it, but any absorbed light, visible or microwave, becomes heat.
- Thermal cameras read 8–14 µm emission. Glass and walls are opaque there, and low-emissivity shiny surfaces mostly show reflections. Kirchhoff's law: good absorbers are good emitters.
- Greenhouse gases (CO_2, H_2O) pass sunlight but absorb Earth's outgoing infrared: about 255 K without them, about 288 K with them.
- Microwave ovens heat by dielectric relaxation of water (a broad hump near 20 GHz), not a sharp resonance. Low-ε surfaces keep heat in; high-ε radiators dump heat in space.

PART IV · **RADIATION**

10 Ionizing Radiation, Shielding, and Space

full chapter p. 106

"How does a magnetic field help protect against radiation?" p. 114

A magnetic field deflects *charged* particles (protons, electrons, alpha particles, cosmic-ray nuclei) so they spiral along field lines instead of reaching the surface, but it does nothing at all to photons (UV, X-rays, gamma rays) or neutrons.

KEY IDEA

What matters is the energy per packet, not the total energy. A beam of a trillion weak photons can warm you, but it cannot ionize. One energetic photon can.

KEY IDEA

"Non-ionizing" does not mean "harmless." Visible and IR photons are too weak to break most bonds, but UV photons (3–10 eV) are in the bond-breaking range even though they cannot ionize.

KEY IDEA

A charged particle tugs on every electron it passes, so it loses energy steadily and stops at a definite **range**. A photon or neutron travels untouched until one chance collision, so photons fall off exponentially (Section 10.8) instead of stopping at a wall.

KEY IDEA

A gray is a tiny amount of heat: 1 J/kg warms water by 0.00024 °C. A sudden whole-body 4–5 Sv kills half of those exposed, yet warms the body by about a thousandth of a degree. What matters is *where* the energy goes (DNA bonds), not how much.

SUMMARY

- Ionizing means about 10 eV or more per photon or particle ($\lambda \lesssim 124$ nm). Brightness cannot substitute for photon energy.
- UV-B is below the threshold but damages DNA by photochemistry (thymine dimers): sunburn.
- Alpha: stopped by paper or skin. Beta: by millimeters of aluminum or plastic. Gamma/X-rays: reduced exponentially, best by dense mass. Neutrons: slowed by hydrogen.
- DNA is damaged directly or by •OH radicals; double-strand and clustered breaks are the worst.
- Gray = J/kg. Sievert = gray × biological weighting. The US average is about 6.2 mSv per year, half of it natural.
- Inverse-square law: $I \propto 1/r^2$. Attenuation: $I = I_0 e^{-\mu x}$, HVL $= 0.693/\mu$. For MeV gamma rays, mass per area is what counts.
- Magnetic fields bend charged particles into helices ($r = mv/qB$) and do nothing to photons or neutrons. Earth relies on both its magnetosphere and its atmosphere (about 10 m of water equivalent).

11 X-Rays and Materials Science

full chapter p. 117

"What does 'X-ray vision' mean — how do X-rays see through things?" p. 119

X-rays don't "see" anything: an X-ray image is a **shadow**.

"How can parallel planes have different Miller indices — what's the deal with (110) vs (220), or (100) vs (200)?" p. 123

(220) planes have the same orientation as (110) planes but are packed twice as closely: $d_{220} = d_{110}/2$.

"If the interplanar spacing is higher, does that mean the density is higher?" p. 124

Within one crystal, yes for the *planar* density: the families of planes with the largest spacing are the most densely packed planes.

KEY IDEA

Diffraction needs a wavelength comparable to the spacing being probed (Chapter 8). Atoms are about 0.1–0.4 nm apart, so the probe must be X-rays (or electrons or neutrons) with wavelengths of a fraction of a nanometer.

KEY IDEA

A Bragg "reflection" is not a mirror bouncing light. It is interference: the planes scatter X-rays at every angle, but the scattered waves add up only at angles where $2d\sin\theta = n\lambda$. A diffractometer swings its detector through the angle 2θ between the incoming and outgoing beams, so patterns are plotted against 2θ.

SUMMARY

- X-ray tubes: bremsstrahlung gives a continuum down to $\lambda_{\text{min}} = hc/eV$; K-shell vacancies give element-specific Kα/Kβ lines (Cu Kα = 0.1542 nm).
- X-ray images are shadows: photoelectric absorption scales roughly as Z^3–Z^4/E^3, so bone and metal absorb far more than soft tissue. CT maps μ in 3D.
- Bragg's law $n\lambda = 2d\sin\theta$, with θ from the planes; patterns are plotted against 2θ.
- Cubic spacing: $d_{hkl} = a/\sqrt{h^2+k^2+l^2}$. (220) planes are parallel to (110) at half the spacing, and a (220) reflection is the second order of (110).

- BCC needs $h + k + l$ even; FCC needs all odd or all even. (100) vanishes in both because halfway-plane atoms scatter exactly out of step.
- Planar density = $n_v \times d$: the widest-spaced planes are the densest (FCC {111}, BCC {110}), and they are also the slip planes.
- 100 kV electrons have λ ≈ 0.0037 nm, enabling TEM, electron diffraction, and EDS.